NAGAPUJITHA KARTHIKEYAN

COMPUTER SCIENCE UNDERGRADUATE (HONORS) | ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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PROFESSIONAL SUMMARY

Computer Science undergraduate with a 3.93/4.0 CGPA and a strong record of academic distinction, now transferring into the Computer Science program at the University of Massachusetts Amherst. Hands-on experience building deep learning and computer vision models (TensorFlow/Keras), supported by a remote AI internship and a foundational data science certification from IIT Madras. Combines technical depth with proven campus leadership, and is seeking part-time opportunities to further develop skills in AI/ML while contributing to real project work. Keen on gaining further knowledge in the AI domain.

EDUCATION

University of Massachusetts Amherst *2026 – 2028*
BS in Computer Science (Transfer, Undergraduate)

ICAS, Manipal Academy of Higher Education (MAHE) *2024 – 2026*
B.Sc. Computer Science (Undergraduate) | Cumulative CGPA: 3.93/4.0

Global Indian International School *2020 – 2024*
Grades 9–12 | CBSE | 92.6% (Grade 12), 94.8% (Grade 10)

PROFESSIONAL EXPERIENCE

Artificial Intelligence Intern — 1Stop.ai (in partnership with Fox Trading Solutions) *Summer, Remote*

- Completed a remote summer internship focused on Artificial Intelligence, gaining applied exposure to AI/ML concepts, tools, and workflows in a professional setting.
- Collaborated remotely as part of a structured internship program, translating AI/ML concepts into practical project work under mentorship.

PROJECTS

Indian Monuments Image Classifier

- Built a multi-class deep learning classifier to recognize Indian heritage monuments from images, using transfer learning on a pre-trained ResNet50V2 backbone (Keras/TensorFlow).
- Fine-tuned the network's deeper layers on 299×299 images with a custom data augmentation pipeline (flips, rotation, zoom, contrast, translation) and L2-regularized output layer to reduce overfitting.
- Applied early stopping and learning-rate scheduling callbacks, and visualized training/validation accuracy and loss curves to evaluate model performance.

Pet Breed Image Classifier

- Designed an end-to-end image classification pipeline for pet breed recognition from a flat, filename-labeled dataset, including an automated label extraction and stratified train/validation split (scikit-learn).
- Fine-tuned a ResNet50V2 model with a custom TensorFlow data pipeline, on-the-fly augmentation, and model checkpointing to automatically retain the best-performing weights during training.

White Blood Cell Detection & Localization

- Developed a multi-task computer vision model that simultaneously classifies and localizes white blood cells in medical microscopy images, parsing JSON-based bounding box annotations into a custom TensorFlow data pipeline.
- Designed a dual-headed MobileNetV2 architecture with separate classification and bounding-box regression heads, implementing a custom IoU metric and visualizing predicted vs. ground-truth boxes to assess accuracy.

Online Doctor Appointment System

- Developed a complete web-based platform to streamline appointment booking and scheduling for clinics using XAMPP (Apache, MySQL), demonstrating full-stack development fundamentals.
- Designed and implemented the underlying MySQL database structure for efficient management of user registration, doctor credentials, and appointment scheduling logic.

CERTIFICATIONS

- Data Science and Applications (Foundation Level) — **IIT Madras**
- Artificial Intelligence Internship Completion Certificate — **1Stop.ai**, in partnership with Fox Trading Solutions
- Project Completion Certificate — **1Stop.ai**, curriculum developed in partnership with Microsoft

UNDERGRADUATE ACHIEVEMENTS

- Rank 1 (Class Topper), Semester 4 — **3.93 CGPA** (out of 45 students)
- Rank 1 (Class Topper), Semester 3 — **3.91 CGPA** (out of 45 students)
- 2nd Class Topper, Semester 2 — **3.86 CGPA** (out of 45 students)
- 3rd Class Topper, Semester 1 — **3.72 CGPA** (out of 45 students)
- **Best Outgoing Student of the Batch**

DESIGNATIONS / ROLES

- **Foreign University Facilitation Secretary** — ICAS, MAHE (Undergraduate Year 2)
- **Class Representative** — ICAS, MAHE (Undergraduate Year 1)
- **Member, Research Division** — Enigma Technical Society

PRE-UNIVERSITY ACHIEVEMENTS

- Awarded Scholarship in Grade 11 and Grade 12 for academic excellence.
- Achieved **International Rank 9** in the International English Olympiad (IEO).
- Recognised as **Biology Topper** and Second Place Overall Topper, Grade 11.

SKILLS

- **Programming:** C++, Java, Python
- **AI / Machine Learning:** TensorFlow, Keras, scikit-learn, OpenCV, Computer Vision, Transfer Learning
- **Web & Database:** SQL Server, MySQL, XAMPP (Apache/MySQL deployment)
- **Tools:** Data Science toolkit (Foundation Level), specialized editing software
- **Soft Skills:** Leadership, Communication, Creativity, Diligence, Proactiveness, Punctuality

LANGUAGES

English | Tamil | Telugu | Hindi